Project Synopsis

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Project Title:
Adaptive Traffic Signal Control Using Traffic-Demand and Waiting-Time Analysis

Brief description:
Urban traffic congestion is often aggravated by fixed-time traffic signals that allocate predetermined green-light durations without considering the actual traffic conditions at an intersection. This project proposes a software-based adaptive traffic signal control system that dynamically adjusts signal phases according to the traffic demand and waiting conditions of vehicles approaching a signalized urban intersection.

The system will be developed and evaluated using Simulation of Urban Mobility (SUMO) as a controlled traffic simulation environment, with Python as the primary programming language and TraCI/libsumo for communication between the control algorithm and the simulation. The controller will monitor the traffic state of configurable intersection approaches, determine their relative priorities, and dynamically allocate green-light durations based on traffic demand and waiting conditions.

A key feature of the proposed system is the prevention of starvation of low-traffic approaches. Instead of prioritizing an approach solely according to vehicle density, the controller will incorporate waiting-time and fairness considerations so that vehicles on less congested approaches are not indefinitely delayed. The adaptive controller will be evaluated against a conventional fixed-time signal controller under identical traffic conditions using metrics such as average waiting time, queue length, travel time, and vehicle throughput.

The core implementation will establish the control algorithm and its performance through simulation, providing a foundation for subsequent integration with real-time traffic data and physical traffic-signal infrastructure.

Objective/s:

1. To develop a configurable model of a signalized urban intersection capable of representing varying numbers of traffic approaches and changing traffic demand.
2. To design an adaptive signal-control mechanism that dynamically determines signal priorities and green-light durations based on the prevailing traffic state.
3. To incorporate waiting-time and fairness considerations into signal decisions to prevent continuous neglect of low-demand traffic approaches.
4. To establish real-time interaction between the control algorithm and the traffic simulation environment, enabling dynamic modification of signal phases during simulation.
5. To evaluate the adaptive control strategy under diverse traffic conditions using relevant traffic-flow, congestion, and fairness-related performance metrics.
6. To compare adaptive and conventional fixed-time signal control using identical traffic scenarios and determine the effectiveness and limitations of the proposed approach.

Software requirements:

- ***Programming Language*:** Python
- ***Traffic Simulation*:** Simulation of Urban Mobility (SUMO)
- ***SUMO-Python Communication*:** TraCI / libsumo
- ***Numerical and Data Processing*:** NumPy, Pandas
- ***Data Visualization*:** Matplotlib
- ***Development Environment*:** Visual Studio Code / PyCharm
- ***Version Control*:** Git and GitHub

Hardware Requirements:

No dedicated hardware is required for the core phase of the project. A standard computer system capable of running Python and SUMO will be sufficient for development, simulation, experimentation, and performance evaluation.

The system will, however, be designed with future real-world integration in mind. In subsequent development phases, real-time traffic information may be obtained through existing CCTV cameras or suitable traffic sensors, while the adaptive controller may be interfaced with authorized traffic-signal control infrastructure. This hardware integration is currently treated as a future extension.

Internal Guide

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